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GitHub Fundamentals

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GitHub – Git.

- Github is a platform widely by developers for collaborations and sharing.
  - Place to store Code, Projects, Artifacts
  - track changes
  - Version Control
  - Pull requests (Review before merging)

Widely used in CI/CD to push code changes and initiate trigger pipeline – automated build, test, deploy.

Git (Version control tool, Locally installed on your computer or server to track code changes and commit.)

GitHub (Online platform for hosting Git repos. access via API, Website – Supports collaborations but network access).

- Repositories: Centralized digital Storage location where we can manage, track code, data & documents. Enables collaborations without overriding

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Local Repo.
— Repo stored in own PC.
— Edit, Stage changes, Commit them.

Remote Repo
— Stored in Remote Server (GitHub)
— Share code and collab - Synchronise

Concepts:

a. Staging: Selecting file changes (add) commit
Staging area — index.
b/w Working & local

b. "git status": Show current status of working directory, staged area modified files, ... etc

c. "git add": Move selected files to staging area — ready for commit.

Syntax: git add app.py / . / *.py
app.py → file
. → all change

d. git commit: Saves changes to local repo — read to be pushed → Remote

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e. "git init" : initializes empty, new, reinitialize repos.
creates hidden .git folder containing metadata, configuration → to track project history.

Git Push –

Syntax: git push Origin main / feature-branch

(R.RP Loc.)

task: Upload local commits to a remote repo....

Git Pull –

Syntax: git pull origin main

task: downloads changes from a RRp

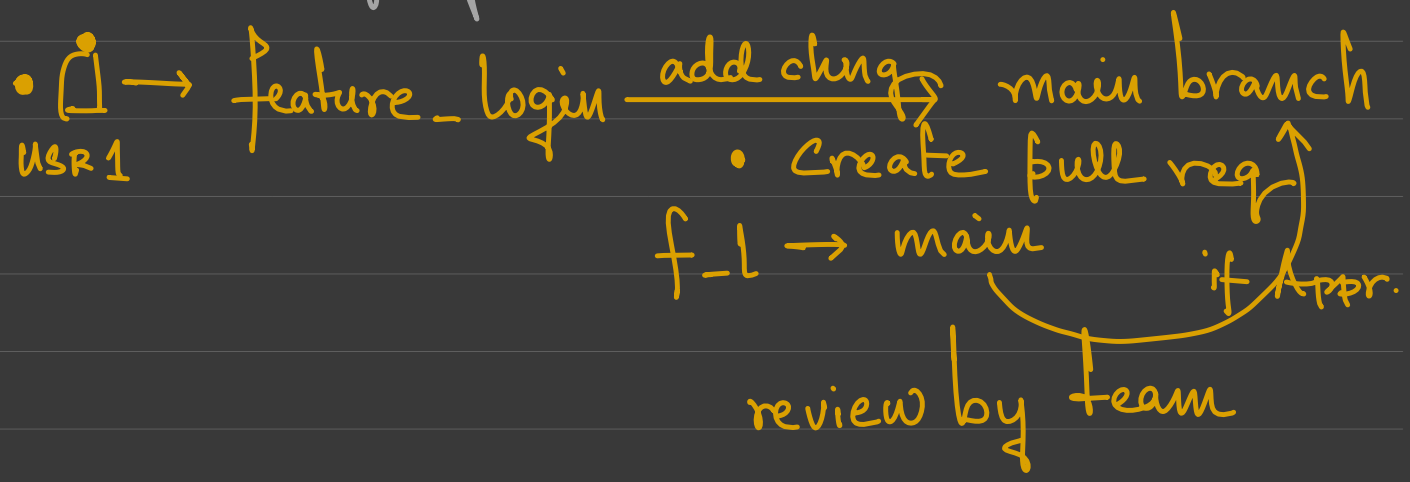
integrate my current local branch

Cycle. fetch → integrate → merge

Git fetch: downloads changes for review doesn't integrate (merge) with local repo.

PR (Pull requests):

feature which enables other developers to review and approve our changes before merging one branch → another.



Git Branch: Branch in github Separate Line of development which lets dev work on a feature or bug fix without directly changing main code.

main (shopping)
- login & Acc
- UI / dashboard
- products / inventory
- payments
- order & shipping

git branch
- list local Br

git branch dev
- create Br

git branch -a
- list local + Remote Br

Git Switch / checkout: Used for switching branches (restore file to prev. state)

checkout -b <name> / switch -c <name>

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Workflow of cmds: in Corporate

<main>

- git pull origin main

<local>

- git switch -c feature/payments

build ... test ... validate

(locally)

- git add . (Stage changes for commit)
- git commit -m "done" (Saved locally)
- git push -u origin feature/payment

<Remote>

github → PR → Main

Pull Request
- Validations
- checks

(if approved) main

Branch Protection: Rules within production repo to prevent accidental push to main

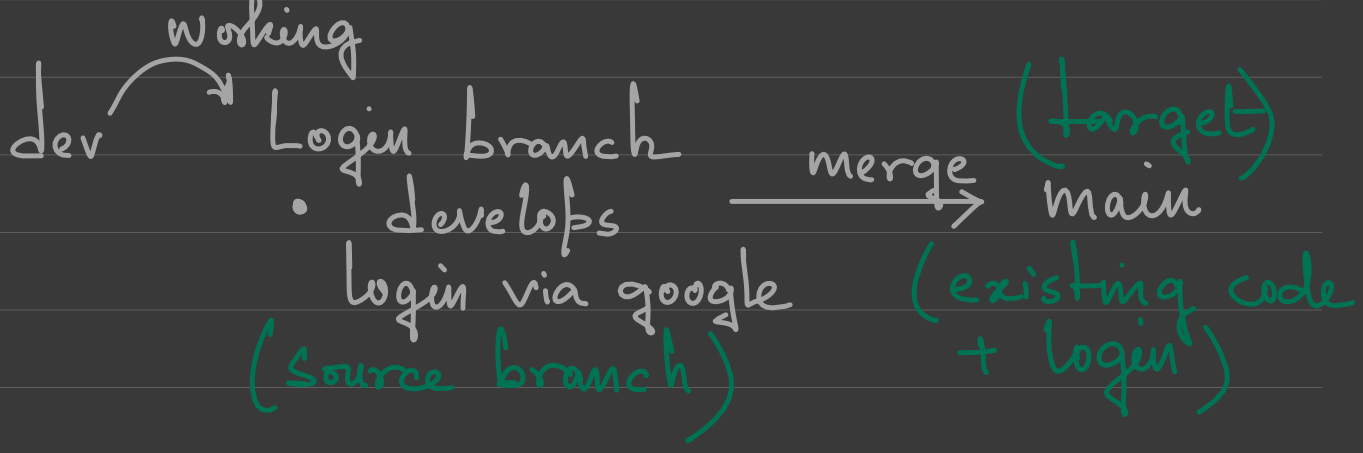
- No direct push
- PR required
- CI must pass
- Code review required

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Merge, Merge Actions, Merge Conflict

Merge : Combines features & changes from one branch to another

Eg:



Working :

git branch , git switch main, git pull origin main

git merge feature/Login , git push origin main

Merge Actions: merging a pull Req after reviews are complete.

- Merge commit: Combines branches, preserving history
- Squash Merge: Combines all PR → 1 commit and merge → target
- Rebase Merge: Updating your code of branch (code) → before merging ..... prevents

messy history

Merge-Conflict occurs when more than one developer is working on the same file and gitHub cannot decide which changes to keep while Combining branches.

- Main : app.txt → "Devops"
- feature : " → "Aws Devops"

conflict

Resolve: → Attemp merge, review, decide correct version then status, add, commit, push.

Git Clone, Rebase:

Git Clone: Used to copy a repository from gitHub or remote to the local machine. — files, commit, history and branches.

Git rebase: moves branch's commit on top of latest commits from another branch — creating more linear history

M: A-B-C
F: A-B-D-E → Rebase: (A-B-C-D-E) (FB)

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- **.gitignore:** Explicitly tells git which files should not be tracked or uploaded.
  .env, .log .... etc files with secret info must never be pushed to remote repo.

git reset: moves head backward to undo commits, modify, rewriting commit

git revert: Creates a new commit that reverses a previous commit → No alter to history

git history: Views chronological record of project.

git stashing: Temporary shelving of uncommitted changes

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